

A-6-B

Roll No.

Total No. of Questions : 29]

[Total No. of Printed Pages : 8

12thSZARJD22

6006-B

MATHEMATICS

Time : 2.30 Hours]

[Maximum Marks : 100

Section-A

(Multiple Choice Questions)

1 eac

1. The relation R in the set {1, 2, 3} given by $R = \{(x, y) \mid x > y, x, y \in A\}$ is :

(A) Reflexive

(B) Symmetric

(C) Transitive

(D) None of these

2. The principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is :

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{6}$

(C) $\frac{2\pi}{3}$

(D) $\frac{\pi}{4}$

Turn O

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8. Find the rate of change of the area of a circle with respect to its radius r when $r = 4$ cm.

9. Find the angle between the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$.

10. Solve the following L.P.P. graphically :

Maximise :

$$Z = 3x + 4y$$

Subject to the constraints :

$$x + y \leq 4,$$

$$x \geq 0, y \geq 0$$

11. If $P(A) = \frac{6}{11}$, $P(B) = \frac{5}{11}$ and $P(A \cup B) = \frac{7}{11}$, find $P\left(\frac{A}{B}\right)$.

12. Given two independent events A and B such that $P(A) = .3$,

$P(B) = .6$. Find $P(A \text{ or } B)$.

Section-C**(Short Answer Type Questions)**

4 each

13. Given $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 4x + 3$. Show that f is invertible.

Find the inverse of f .

14. Write the following function in the simplest form :

$$\tan^{-1} \left[\frac{\cos x - \sin x}{\cos x + \sin x} \right] \quad 0 < x < \pi$$

15. Express the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ as the sum of symmetric and

a skew-symmetric matrix.

16. Find the relationship between a and b so that the function f defined by :

$$f(x) = \begin{cases} ax+1 & \text{if } x \leq 3 \\ bx+3 & \text{if } x > 3 \end{cases}$$

is continuous at $x = 3$.

17. Find the intervals in which the function f given by $f(x) = 4x^3 - 6x^2 - 72x + 30$ is :

(a) Strictly increasing

(b) Strictly decreasing

18. Find the equations of the tangent and normal to the given curve at the indicated point $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(0, 5)$.

19. Find the general solution of the differential equation :

$$\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$$

20. Find the area of the region bounded by the ellipse :

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

21. Find $|\vec{a} \times \vec{b}|$ if $\vec{a} = \hat{i} - 7\hat{j} + 7\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}$.

22. If $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, then show that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular.

23. Solve the following problem graphically :

Minimise and maximise : $Z = 3x + 9y$

Subject to the linear constraints :

$$x + 3y \leq 60, x \leq y,$$

and

$$x + y \geq 10, x \geq 0, y \geq 0$$

Turn Over

Section-D

6 c

(Long Answer Type Questions)

24. By using properties of determinants prove that :

$$\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix} = a^3$$

Or

If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$, then verify that $A(\text{adj } A) = |A|I$. Also find A^{-1}

25. Find $\frac{dy}{dx}$ of the function $x^y + y^x = 1$.

Or

If $y = 3 \cos (\log x) + 4 \sin (\log x)$, show that :

$$x^2 y_2 + x y_1 + y = 0$$

26. Integrate the rational fraction :

$$\frac{2x-3}{(x^2-1)(2x+3)}$$

Or

Using the properties of definite integrals evaluate :

$$\int_{-5}^5 |x+2| dx$$

27. Find the general solution of the differential equation :

$$x \frac{dy}{dx} + 2y = x^2 \log x$$

Or

Show that the differential equation $(x^2 - y^2)dx + 2xydy = 0$ is homogeneous and solve it.

28. Find the equation of the plane through the intersection of the planes $3x - y + 2z - 4 = 0$ and $x + y + z - 2 = 0$ and the point $(2, 2, 1)$.

Or

Find the angle between the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$ and the plane

$$10x + 2y - 11z = 3.$$

- i) Find the probability distribution of number of doublets in three throws of a pair of dice

Or

Two balls are drawn at random with replacement from a box containing 10 black balls and 8 red balls. Find the probability that

- a) Both balls are red
- b) First ball is black and second is red
- c) One of them is black and other is red